

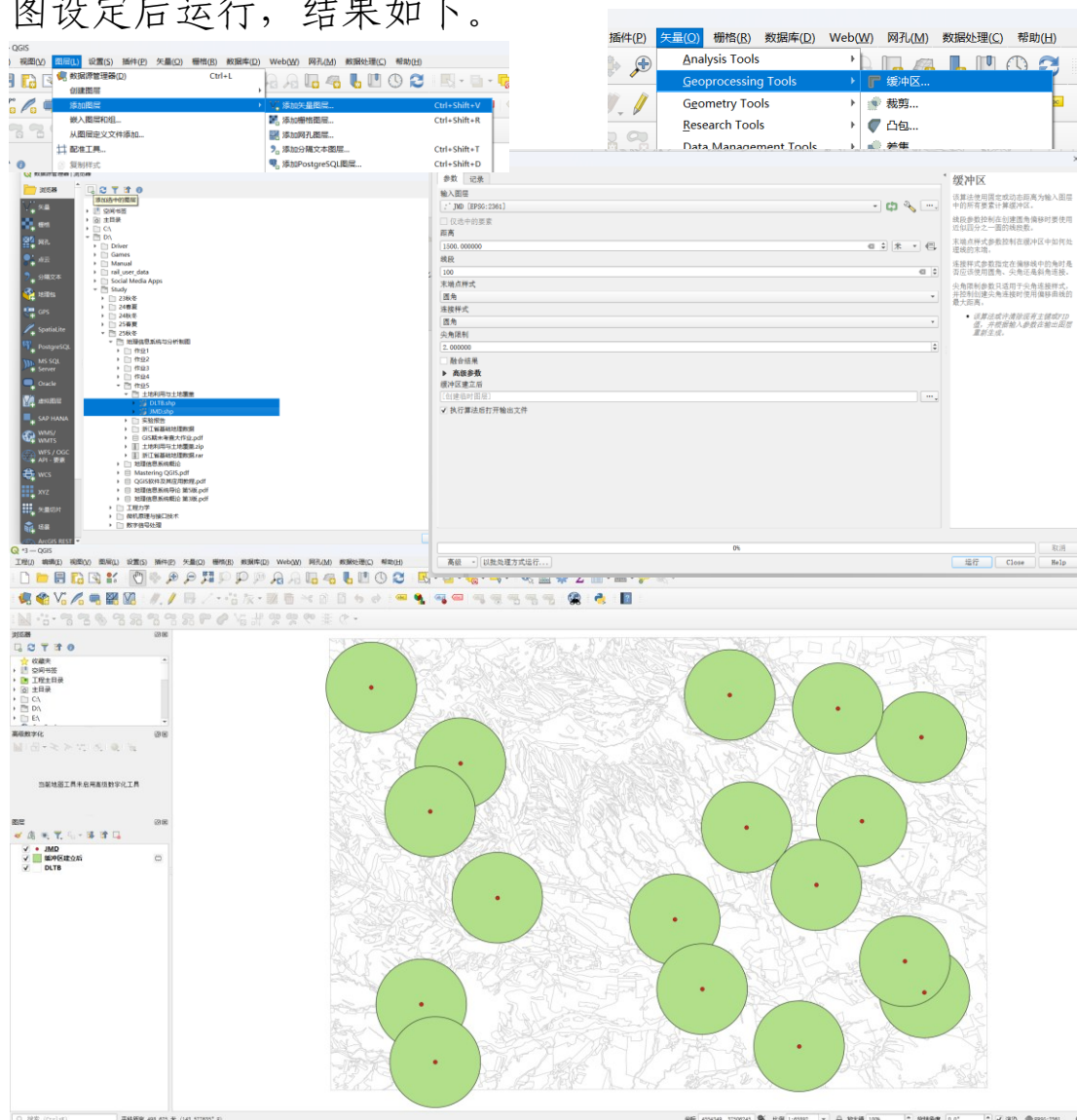
考查大作业 GIS 综合分析与制图-第 3 题

一、实验设备、实验环境

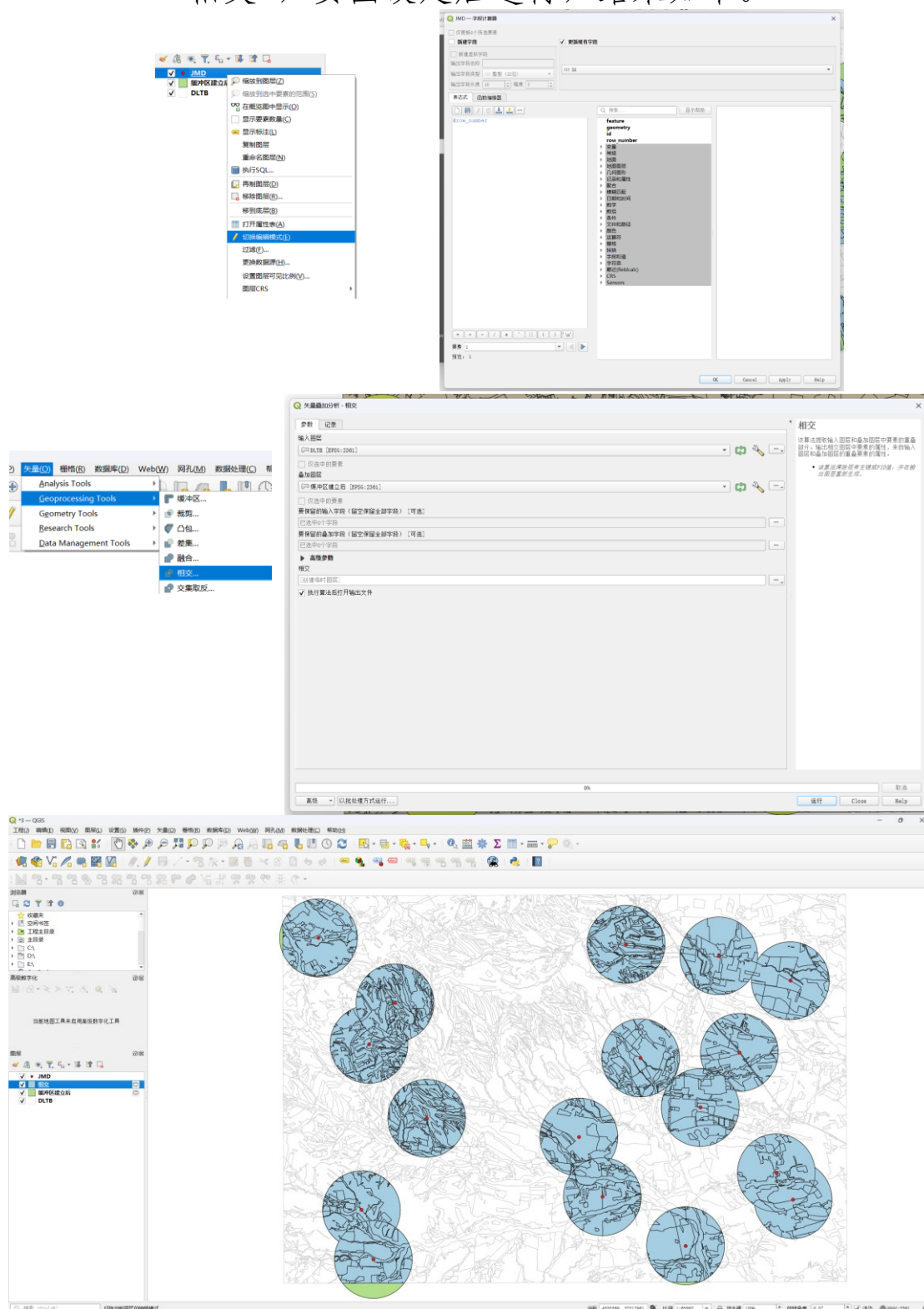
采用 QGIS 3.44 版本的笔记本一台

二、实验步骤

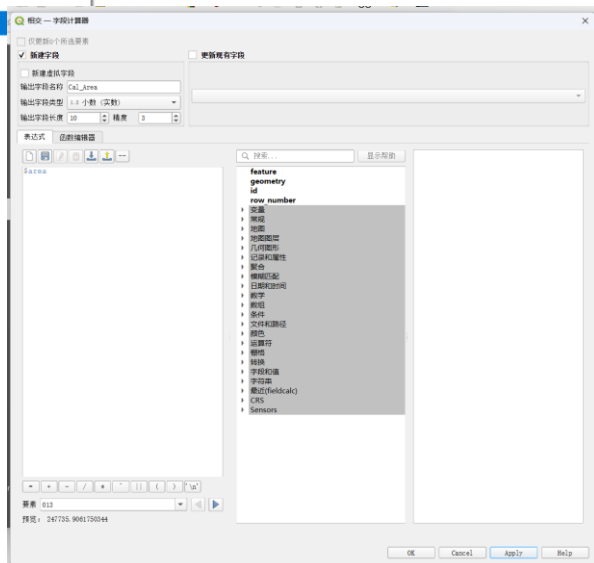
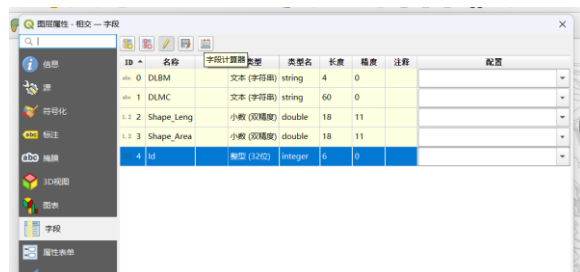
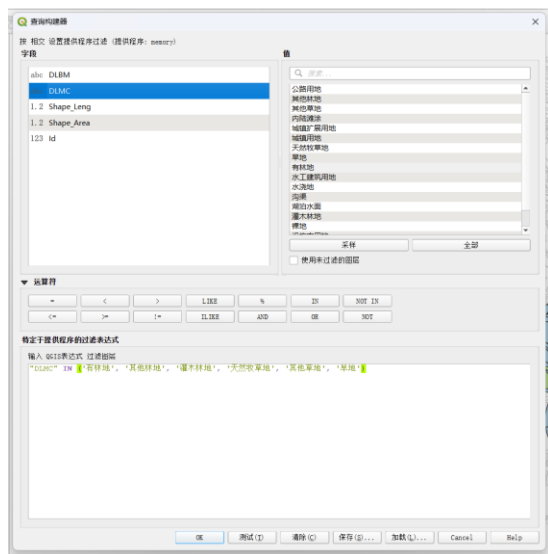
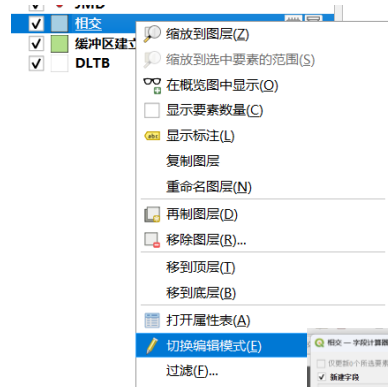
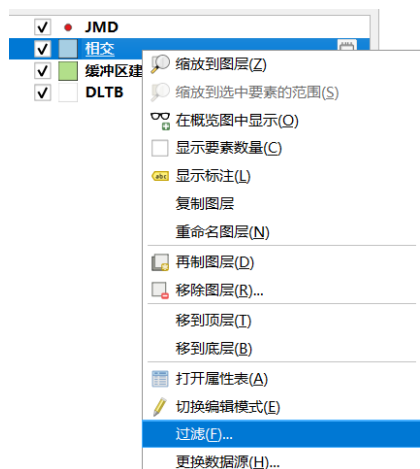
1、打开 QGIS 软件，在菜单栏中选择“图层”-“添加图层”-“添加矢量图层”，在新页面选择浏览器后选中“DLTB.shp”和“JMD.shp”，添加选中的图层。在菜单栏选择“矢量”-“Geoprocessing Tools”-“缓冲区”，按图设定后运行，结果如下。

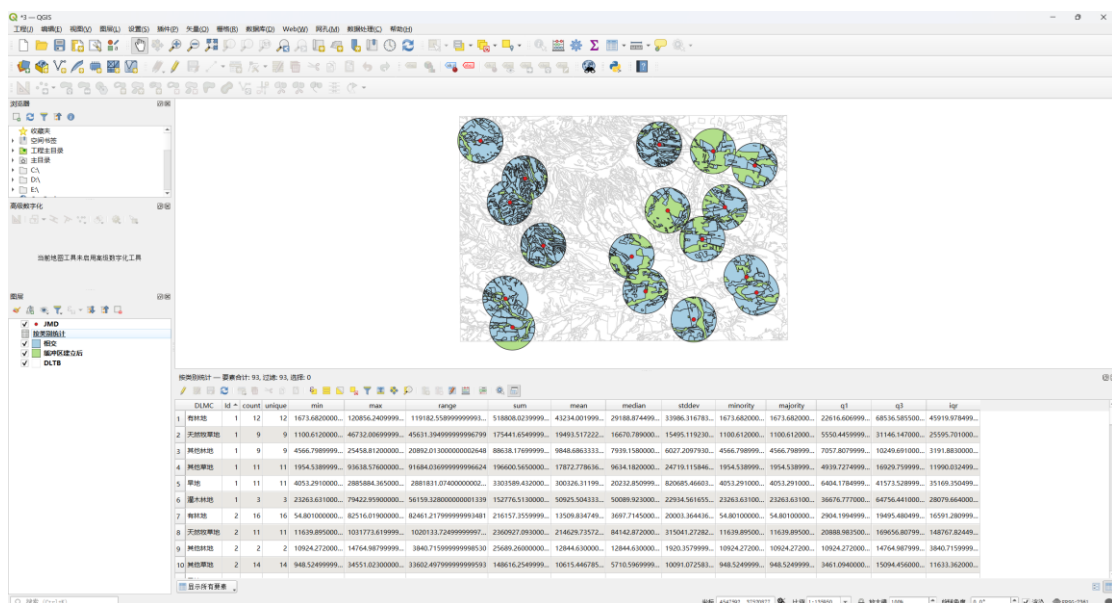


2、右键“JMD” - “切换编辑模式”，在字段计算器勾选“更新现有字段”，选择“id”，表达式输入“@row-number”。菜单栏选择“矢量” - “Geoprocessing Tools” - “相交”，安图设定后运行，结果如下。



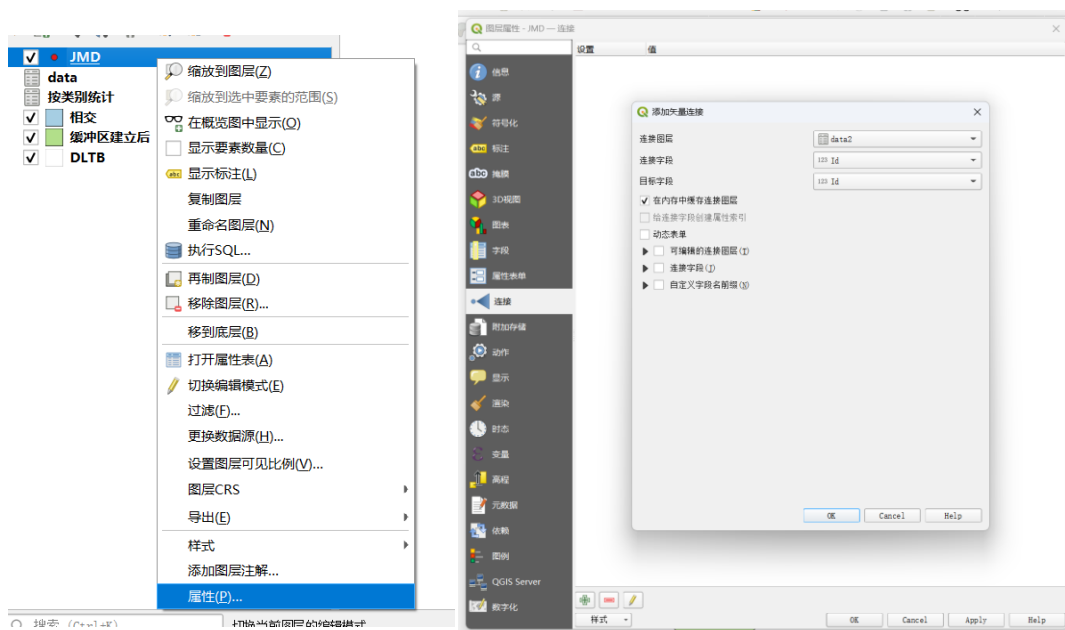
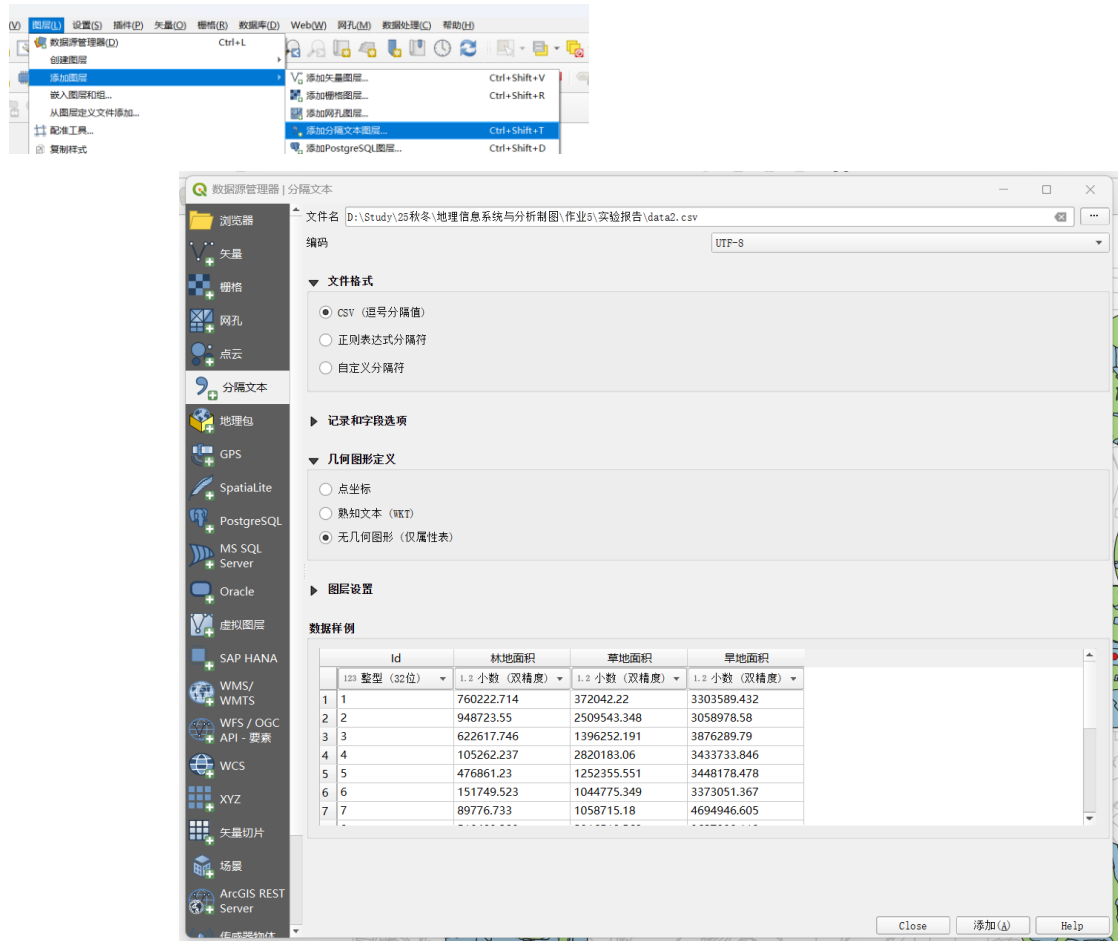
3、右键“相交”-“过滤”，在查询构建器表达式输入“DLMC IN (‘有林地’，‘其他林地’，‘灌木林地’，‘天然牧草地’，‘其他草地’，‘旱地’)”并运行。再右键“相交”-“切换编辑模式”，打开字段计算器后新建字段“Cal_Area”，输出字段类型为“小数”，表达式为“\$area”并保存。



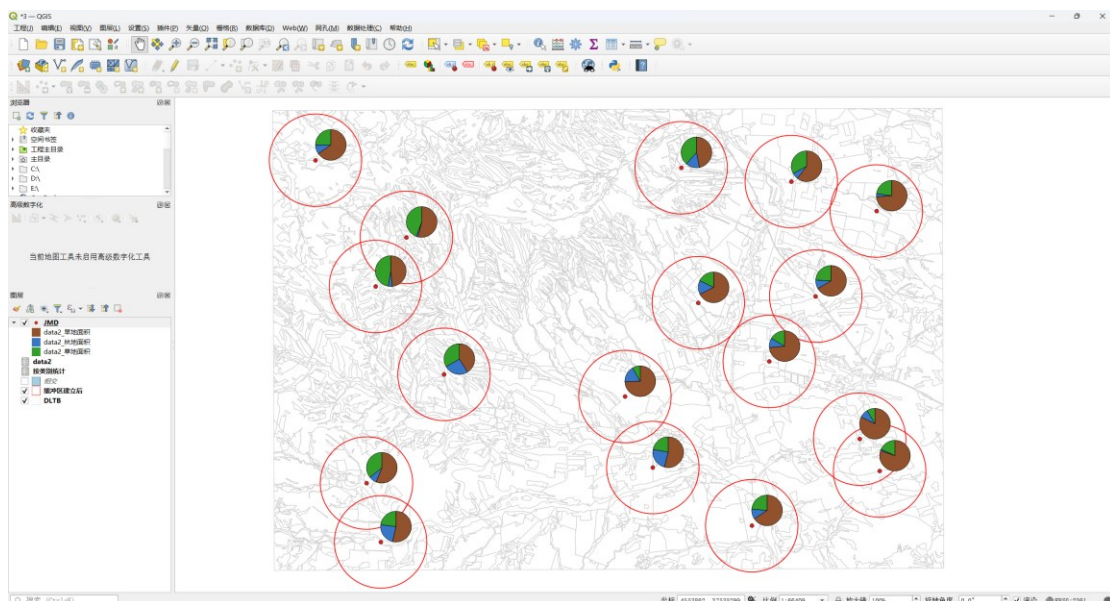
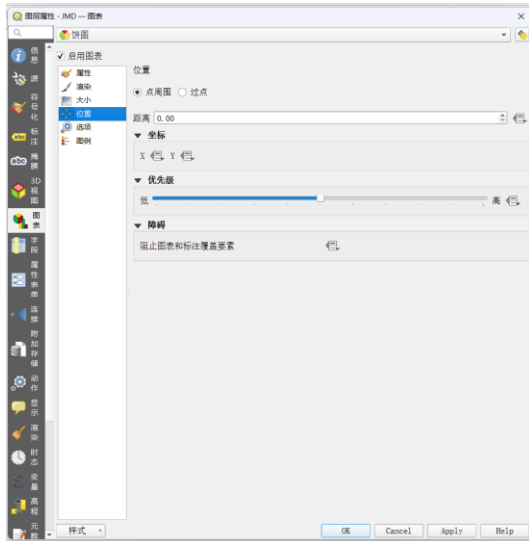
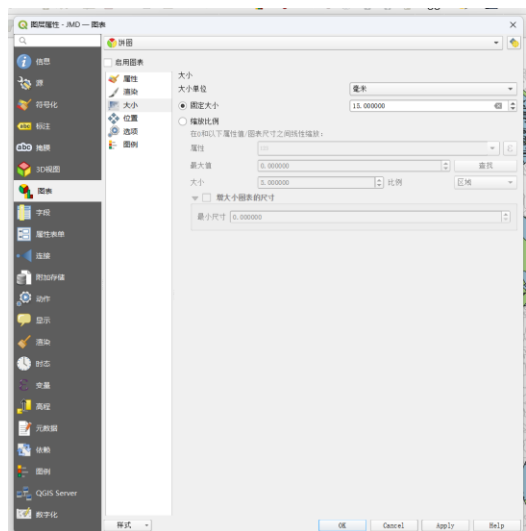
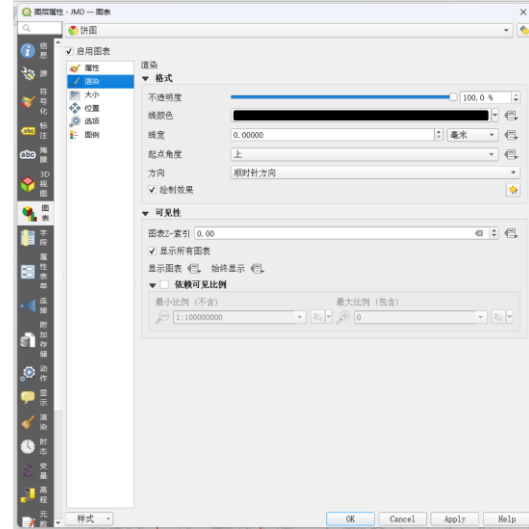
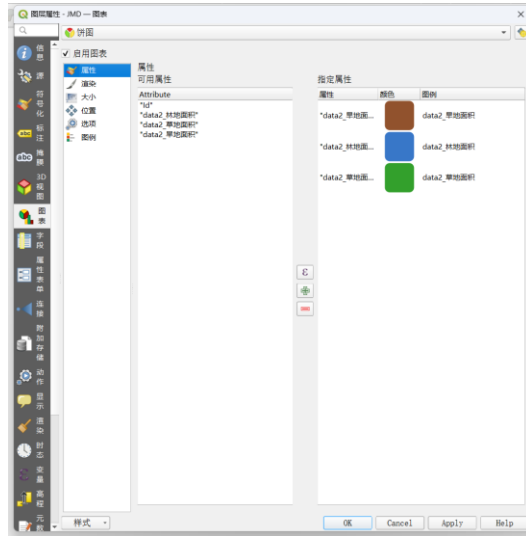


DLMC	Id	count	unique	min	max	range	sum	mean	median	stddev	minority	majority	q1	q3	igr
灌木林地	1	3	3	23263.631	79422.959	56159.328	152776.513	50925.50433	50889.923	22934.56166	23263.631	23263.631	36676.777	64756.441	28079.664
旱地	1	11	11	4053.391	285884.345	285881.074	330336.740	300312.432	4053.391	420688.408	4053.391	4053.391	6404.1765	41573.529	35169.3605
其他草地	1	11	11	1954.539	93638.576	91684.037	106600.565	17872.77864	9634.182	24719.11585	1954.539	1954.539	4939.7275	16929.76	11990.025
其他林地	1	9	9	4566.799	25458.812	20892.013	88638.177	9848.68633	7939.158	6027.209793	4566.799	4566.799	7057.808	10249.691	3191.883
天然牧草地	1	9	9	1100.612	46732.007	45631.395	175441.655	19493.51722	16670.789	15495.11923	1100.612	1100.612	5550.446	31146.147	25595.701
有林地	1	12	12	1673.682	120856.241	119182.559	518808.024	4324.002	29188.8745	33986.1678	1673.682	1673.682	22616.607	68536.5855	45919.9785
灌木林地	2	10	10	1249.447	490528.857	489279.41	706876.934	70687.6934	15133.687	141994.4796	1249.447	1249.447	7162.981	46828.747	39665.766
旱地	2	46	46	346.793	919344.461	918997.652	3058975.58	66499.53435	2231.2775	138748.393	346.793	346.793	9304.841	74306.025	46161.184
其他草地	2	14	14	948.525	34551.023	33602.498	148616.255	10615.44679	5710.597	10091.07258	948.525	948.525	3461.094	15094.456	11633.362
其他林地	2	2	2	10924.272	14764.988	3840.716	25689.26	12844.63	12844.63	1920.358	10924.272	10924.272	10924.272	14764.988	3840.716
天然牧草地	2	11	11	11639.895	1031773.62	1020133.725	2360927.093	214629.7357	84142.872	315041.2728	11639.895	11639.895	20888.9835	169656.808	148767.8245
有林地	2	16	16	54.801	82516.019	82461.218	216157.356	13509.83475	3697.7145	20003.36444	54.801	54.801	2904.1995	19495.4805	16591.281
旱地	3	18	18	345.587	1446025.336	1445179.749	3876289.79	215349.4328	84810.586	343054.3472	345.587	345.587	11185.629	306054.184	294868.555
其他草地	3	4	4	1902.395	103466.625	101564.23	196332.265	49083.06625	4581.6225	36231.77181	1902.395	1902.395	21416.1255	76750.007	55333.8015
天然牧草地	3	19	19	2318.332	447604.766	445376.434	1199919.926	63153.68032	21436.248	104522.9062	2318.332	2318.332	7616.207	61563.75	53947.435
其他林地	3	9	9	2.521	431504.987	431502.466	622617.746	69119.74967	5767.917	132405.9337	2.521	2.521	1212.963	81692.84	78097.877
旱地	4	61	61	1131.795	651589.001	650457.206	3433733.846	56290.71879	16298.235	113800.6229	1131.795	1131.795	7605.928	60450.59	52844.665
其他草地	4	4	4	7210.621	41410.05	34199.429	81375.032	20343.758	16377.1805	12806.61581	7210.621	7210.621	10775.1875	29912.5285	19137.141
天然牧草地	4	35	35	281.529	640201.978	639920.449	2738088.028	78251.65794	15630.193	135897.2418	281.529	281.529	5768.641	81836.218	76067.597
有林地	4	11	11	0.699	26455.68	26454.981	105262.237	9569.29473	5243.38	8926.47322	0.699	0.699	2307.124	16973.348	14466.2235
灌木林地	5	3	3	4665.068	54768.646	50103.578	111893.304	37297.768	52459.59	23094.0507	4665.068	4665.068	28562.329	53614.118	25051.789
旱地	5	24	24	2872.005	1047889.815	1045017.81	3448178.478	143674.1032	45485.366	238345.2496	2872.005	2872.005	10022.4195	166379.4935	156357.074
其他草地	5	16	16	3117.508	136368.739	133251.231	559474.815	34967.17594	20870.867	38875.9439	3117.508	3117.508	10072.415	44937.7965	24865.3815
其他林地	5	5	5	5242.023	54410.83	49168.807	112988.721	22597.7442	14446.423	18674.94951	5242.023	5242.023	6327.023	32562.422	26235.399
天然牧草地	5	12	12	1847.205	28917.449	28733.244	575319.033	47919.0813	3397.311	22017.96758	1847.205	1847.205	1587.2605	17693.344	11992.42
有林地	5	5	5	2016.844	213445.801	211428.957	251979.205	50395.841	12751.131	81663.37389	2016.844	2016.844	7695.212	16070.217	8375.005
灌木林地	6	1	1	16024.359	16024.359	0	16024.359	16024.359	16024.359	0	16024.359	16024.359	16024.359	16024.359	0
旱地	6	21	21	10494.492	964842.113	954347.621	3373051.367	160621.4937	51196.796	239723.5721	10494.492	10494.492	26185.932	166090.9	139904.968
其他草地	6	4	4	11074.177	44559.287	33485.11	102334.787	25583.69675	23350.6615	12049.59068	11074.177	11074.177	17054.915	34112.4785	17057.5635
其他林地	6	2	2	3314.081	5892.556	2578.473	9206.637	4603.3185	1289.2375	3314.081	3314.081	3314.081	5892.556	2578.473	9206.637
天然牧草地	6	2	2	3654.793	6913470.877	6910716.084	134640.562	134640.562	10584.406	232017.0225	3654.793	3654.793	9601.8285	108528.609	95001.409
有林地	6	8	8	1143.645	101089.912	99946.267	126518.527	15814.81588	2704.8751	32377.92182	1143.645	1143.645	1787.6985	7649.915	5862.2165
灌木林地	7	1	1	6767.754	6767.754	0	6767.754	6767.754	6767.754	0	6767.754	6767.754	6767.754	6767.754	0
旱地	7	11	11	76.093	2535464.348	2535388.255	4694946.605	426813.3277	62152.904	790150.5225	76.093	76.093	103096.011	223469.5905	2143171.5795
其他草地	7	5	5	3187.26	216431.376	213244.116	335913.482	67182.6964	33750.614	75904.1118	3187.26	3187.26	36939.113	41785.119	48860.776
其他林地	7	1	1	76841.833	76841.833	0	76841.833	76841.833	76841.833	0	76841.833	76841.833	76841.833	76841.833	0
天然牧草地	7	6	6	40391.08	309388.873	268997.593	722801.698	120466.4907	91958.905	91761.47915	40391.08	40391.08	5401.791	135086.255	81068.315
有林地	7	1	1	6167.146	6167.146	0	6167.146	6167.146	6167.146	0	6167.146	6167.146	6167.146	6167.146	0
旱地	8	18	18	1444.445	1262492.533	1261048.088	3627990.119	201555.0066	89623.6225	338463.6784	1444.445	1444.445	17819.061	171360.537	153541.476
其他草地	8	14	14	884.776	33898.95	33014.174	140253.504	10018.10743	7287.495	9240.635671	884.776	884.776	3637.129	13107.817	9470.688
其他林地	8	2	2	8026.151	53558.272	45532.121	61584.423	30792.2115	30792.2115	22766.0605	8026.151	8026.151	33558.272	45532.121	61584.423
天然牧草地	8	13	13	1714216.783	1714216.783	0	1714216.783	1714216.783	39747.319	489700.9559	6867.778	6867.778	17307.8105	59665.749	40192.9205
有林地	8	15	15	701.792	148861.572	148159.78	456898.86	30459.924	6851.085	42939.18712	701.792	701.792	2702.7805	4738.633	45135.8525
旱地	9	60	60	0.591	645403.856	645403.265	3214352.578	53572.54297	14808.837	104437.8077	0.591	0.591	3802.123	56669.41	52967.289
其他草地	9	8	8	109.767	58589.182	58479.415	167760.928	20970.116	11884.559	19166.7213	109.767	109.767	4670.976	33975.4545	29040.4785
其他林地	9	7	7	8176.644	59722.091	51545.447	174346.891	24919.55586	18994.603	17340.14322	8176.644	8176.644	10849.1355	32922.641	22073.5055
天然牧草地	9	6	6	103.203	768626.332	768626.332	228449.8226	118202.221	24417.3836	2810.937	2810.937	36555.563	442078.691	403426.128	19720.5055
有林地	9	6	6	1548.202	62912.864	61364.662	93916.34	15619.3914	9257.368	1548.202	1548.202	1766.705	17693.344	11992.42	11992.42
旱地	10	24	24	653.262	804468.359	803815.097	1922907.13	80121.13042	14123.333	177564.3465	653.262	653.262	3684.0705	26171.736	22487.6655
其他草地	10	15	15	5054.631	51230.888	50546.631	205574.401	13704.96007	8933.332	12971.59696	5054.631	5054.631	5979.3895	32906.6205	9306.6205
其他林地	10	3	3	4907.918	18254.439	13346.521	40007.236	13335.74533	16844.879	5987.092785	4907.918	4907.918	10876.3985	17549.659	6763.2605
天然牧草地	10	8	8	1102.387	112792.535	111690.148	300477.762	37559.72025	26229.485	33822.71565	1102.387	1102.387	14848.786	52213.149	37364.363
有林地	10	19	19	382.688	101706.001	101323.313	374084.801	19736.4216	4250.238	25186.8087	382.688	382.688	5877.767	10710.7005	16032.9635
灌木林地	11	2	2	71687.351	822021.66	750504.309	394040.013	197004.4655	197004.4655	197004.4655	71687.351	71687.351	1987.525	232021.66	95001.409
其他草地	11	20	20	78.043	2071858.533	2071780.49	4148119.571	207405.986	22500.1185	50385.5888	78.043	78.043	5138.661	107490.9955	100211.3345
其他林地	11	6	6	2220.133	273840.222	271620.089	347011.937	57835.32283	8574.9435	97703.8783	2220.133	2220.133	7457.625	46344.07	38886.445
天然牧草地	11	1	1	30820.482	30820.482	0	30820.482	30820.482	30820.482	0	30820.482	30820.482	30820.482	30820.482	0
其他草地	11	9	9	9248.524	365237.973	355989.449	1258649.324	138949.9249	132506.34	115710.5598	9248.524	9248.524	54274.444	166073.783	111799.339
有林地	11	9	9	1657.943	75465.088	73747.145	196489.712	21165.52556	11318.584	24598.55604	1657.943	1657.943	4423.646	21632.65	17208.819
灌木林地	12	1	1	5376.088	5376.088	0	5376.088	5376.088	5376.088	0	5376.088	5376.088	5376.088	5376.088	0
旱地	12	10	10	10103.951	603090.825	592986.874	1531236.397	153123.6397	28222.5385	195835.4299	10103.951	10103.951	14086.608	292439.716	27883.532
其他草地	12	20	20	426.376	254750.426	254324.05	836233.245	41811.66225	10687.3945	67490.72664	426.376	426.376	4846.3575	25832.8795	

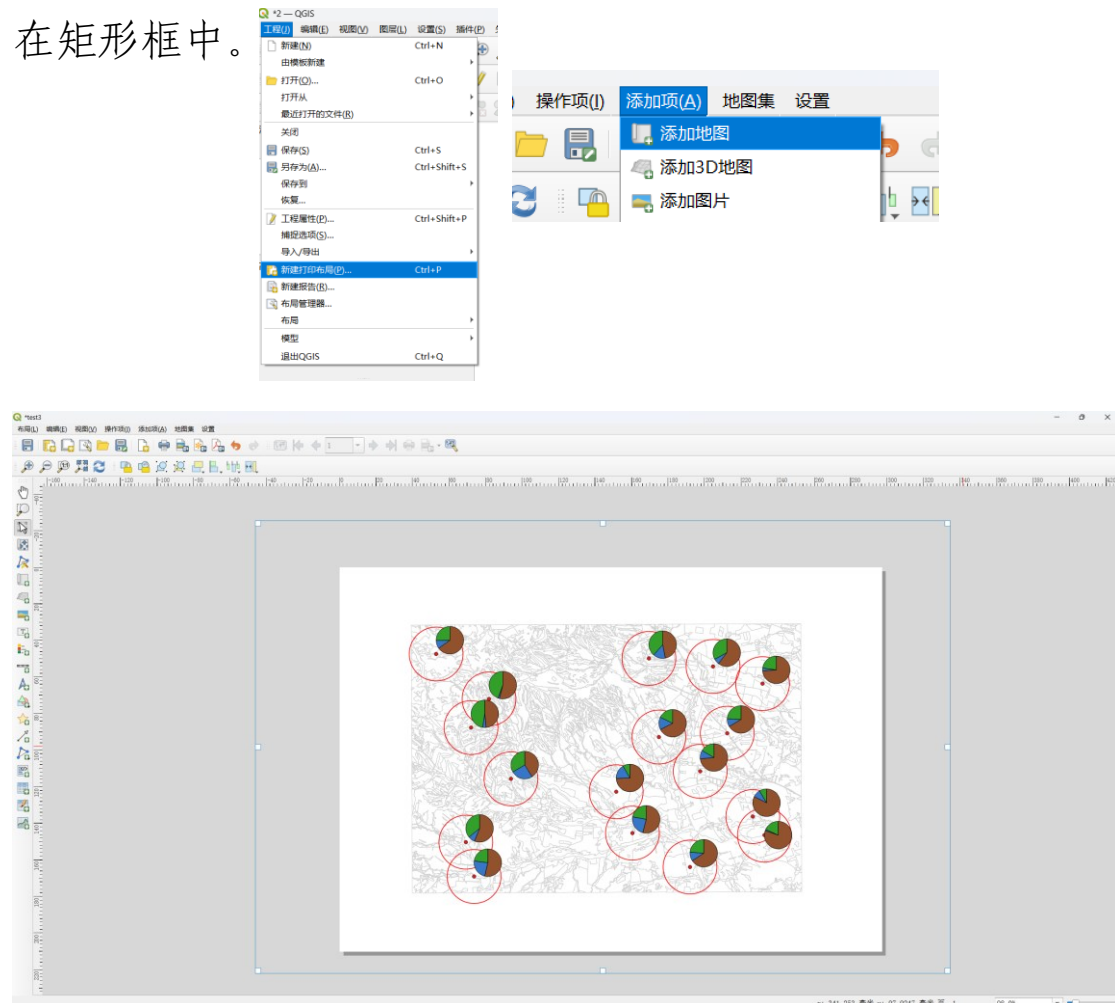
5、在菜单栏中选择“图层”-“添加图层”-“添加分隔文本图层”，导入刚刚的 csv 文件，按图设置后添加。右键“JMD”-“属性”-“连接”，按图设置。



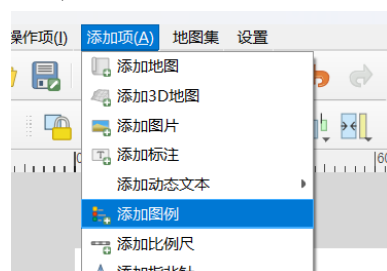
6、右键“JMD”-“属性”-“图表”，选择“饼图”，按图设置后勾选“启用图表”，最终结果如下

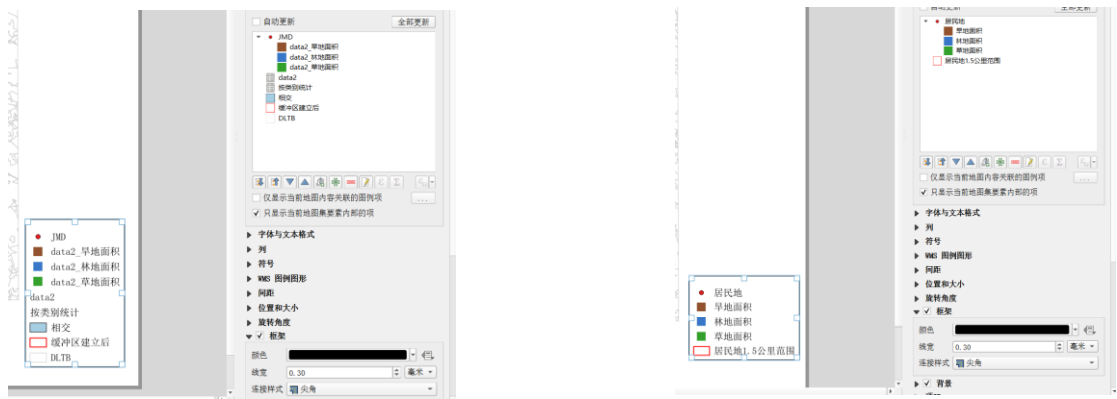


7、在菜单栏中选择“工程”-“新建打印布局”，自定义名称（test3）后会弹出一个空白的白色画布窗口，在工具栏中选择“添加项”-“添加地图”，在白色画布上按住鼠标左键拖拽出一个矩形框，松开鼠标后地图就会呈现在矩形框中。

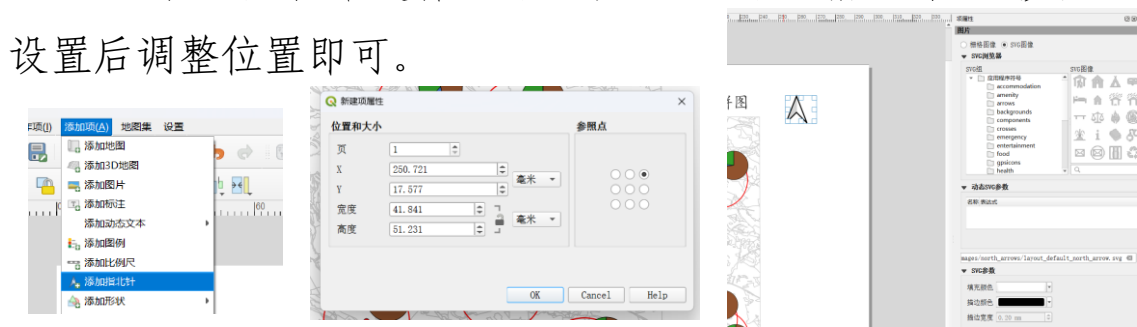


8、在工具栏中选择“添加项”-“添加图例”，取消勾选“自动更新”，勾选“只显示当前地图集要素内部的项”，勾选框架，修改各项数值后结果如图所示。





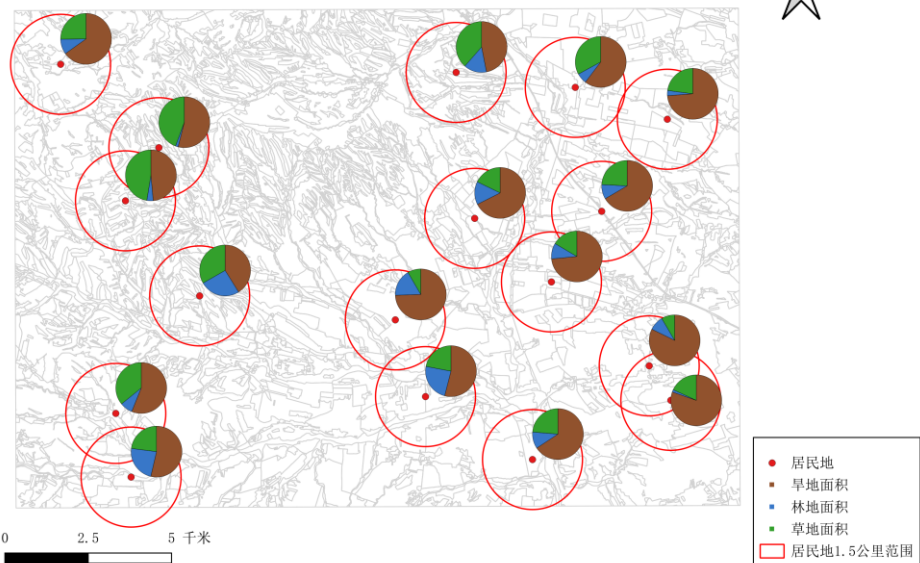
9、在工具栏中选择“添加项” - “添加指北针”，按图设置后调整位置即可。



10、在工具栏中选择“添加项” - “添加比例尺”，按图设置属性。



某居民地1.5公里范围内林地、草地、旱地面积拼图



11、本实验运用以下 GIS 空间分析与数据处理技术：

邻域分析：使用缓冲区构建空间邻域范围。

拓扑修复：清洗原始数据中的无效几何，保证空间运算的可行性。

叠置分析：使用相交提取特定区域内的地理要素。

属性统计与重组：基于空间关系进行分组统计。

专题地图制图：利用统计图表可视化表达属性特征。